

EC-341 Digital System Design

BS(CE)-2021

LAB REPORT # 07



Submitted By:

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	Presentation (Format(0.5)+ Title (0.5)+ Conclusion(1) +Procedure(2) +Objective(1))	Calculation/ Coding	Observation/ Program Results	Total
Total	5	5	5	15
Obtained				

**Lab Instructor.
Engr. Fasih Ahmad**

Experiment #07

Analysis of Implementation of Combinational Circuits Using Different Approaches (Design ALU)

Objective

- Combinational Circuits
- Implementation using ALU
- Comparison of Area Estimations
- Lab Tasks

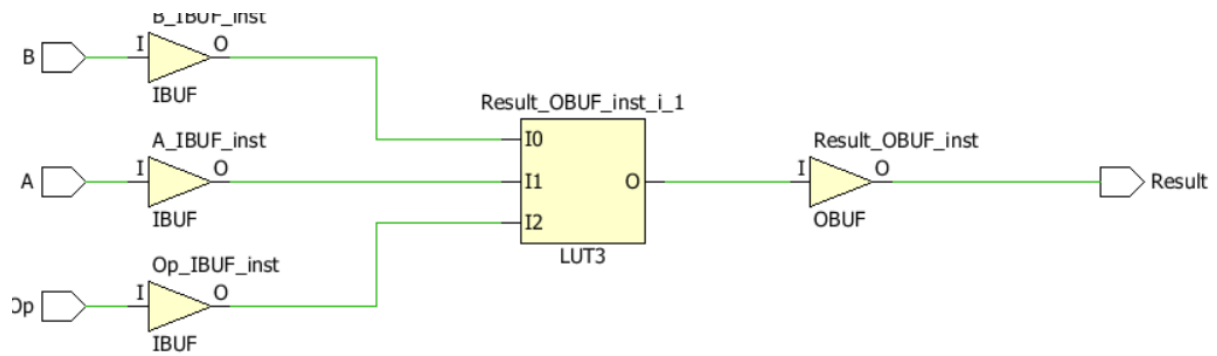
Software Used

- Vivado

Tasks

Task 1

```
23 module ALU(  
24     input A,  
25     input B,  
26     input Op,  
27     output reg Result  
28 );  
29  
30 always @* begin  
31     if (Op == 1'b0) begin  
32         Result = A ^ B;  
33     end  
34     else begin  
35         Result = 1'b0;  
36     end  
37 end  
38 endmodule  
39  
40
```



209 Report Cell Usage:

210	Cell	Count
211		
212		
213	LUT3	1
214	IBUF	3
215	OBUF	1
216		

217

218 Report Instance Areas:

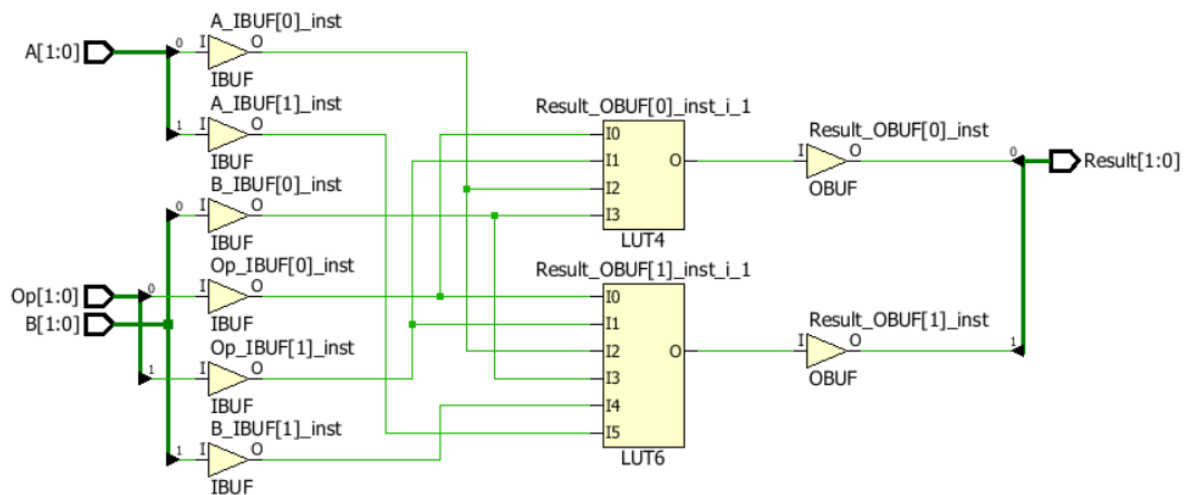
219	Instance	Module	Cells
220			
221			
222	1	top	5
223			

End Point Clock	Requirement	0	1	2	3	4	5	6	7	8	9	10	11-15	16-20	21-25	26-30	31+
	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0

* Columns represents the logic levels per end point clock
 ** distribution is for top worst 1 paths

Task 2

```
module ALU_2bit(  
    input [1:0] A,  
    input [1:0] B,  
    input [1:0] Op,  
    output reg [1:0] Result  
);  
  
always @* begin  
    case(Op)  
        2'b00: Result = A + B;  
        2'b01: Result = A - B;  
        2'b10: Result = A & B;  
        default: Result = 2'b00;  
    endcase  
end  
  
endmodule
```



15 Report Cell Usage:

16	+	-----	+
17		Cell	Count
18	+	-----	+
19		LUT4	1
20		LUT6	1
21		IBUF	6
22		OBUF	2
23	+	-----	+

24

25 Report Instance Areas:

26	+	-----	+				
27			Instance	Module	Cells		
28	+	-----	+				
29		1	top			10	
30	+	-----	+				

31

End Point	Clock	Requirement	0	1	2	3	4	5	6	7	8	9	10	11-15	16-20	21-25	26-30	31+
			0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0

* Columns represents the logic levels per end point clock

** distribution is for top worst 2 paths

Conclusion

In this lab we implemented ALU capable of performing three operations: addition, subtraction, and bitwise AND.